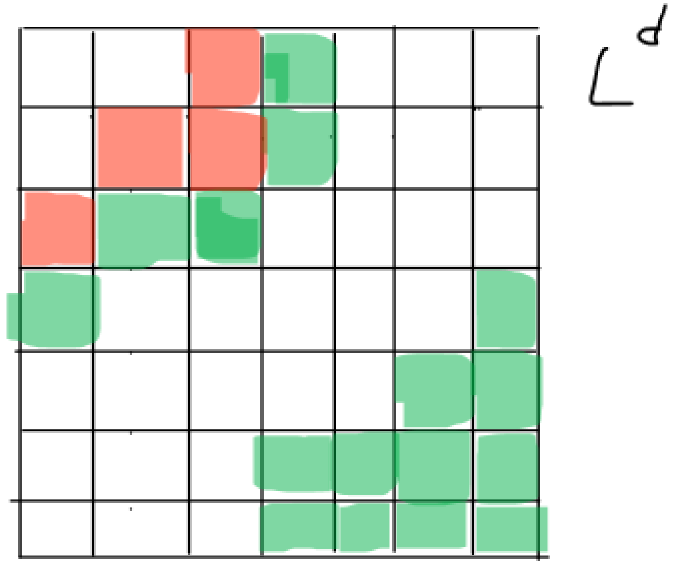
CRITICALLY INFLAMMATORY: A FOREST FIRE MODEL

SET-UP

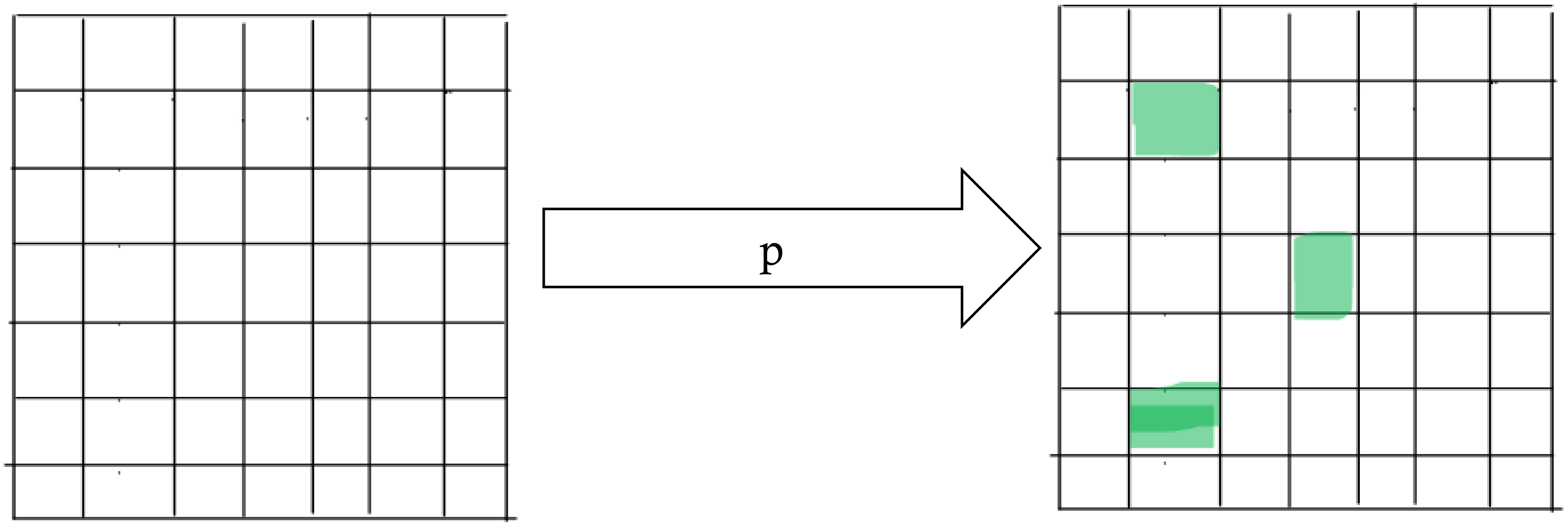
- Lattice
- Each cell can have one of 3 states:
  - Tree (green)
  - Burning (red)
  - Empty (white)



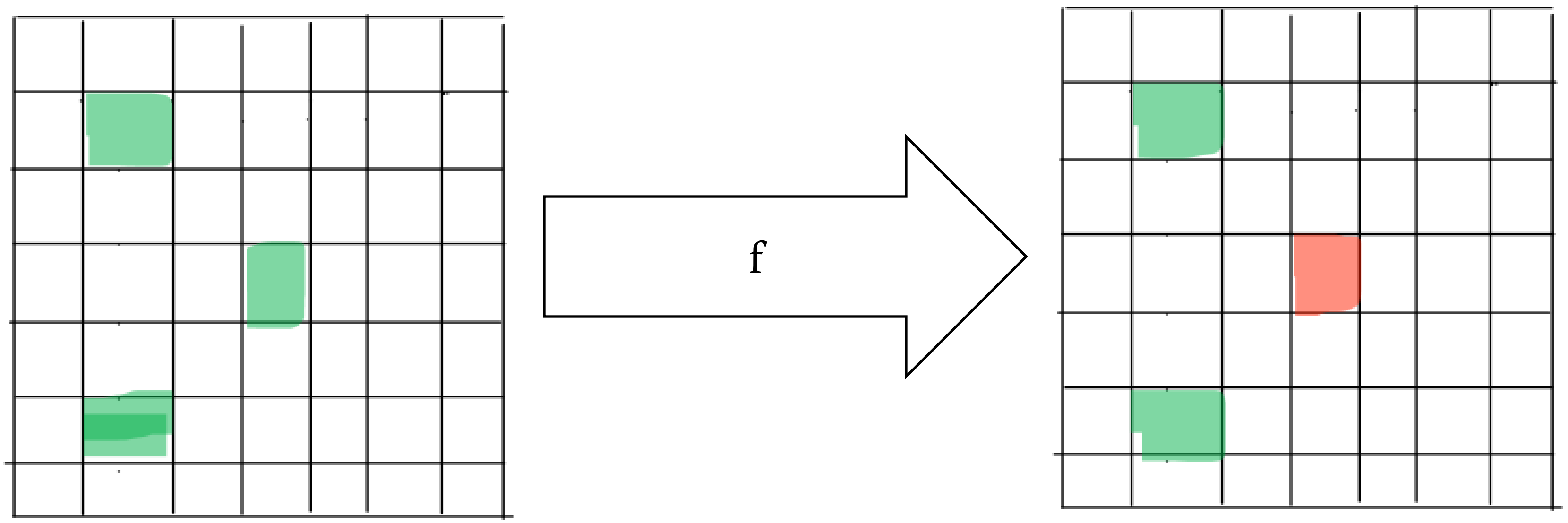
FACTORS:

1. Vegetation rate
2. Lightning rate
3. Ignition Rate

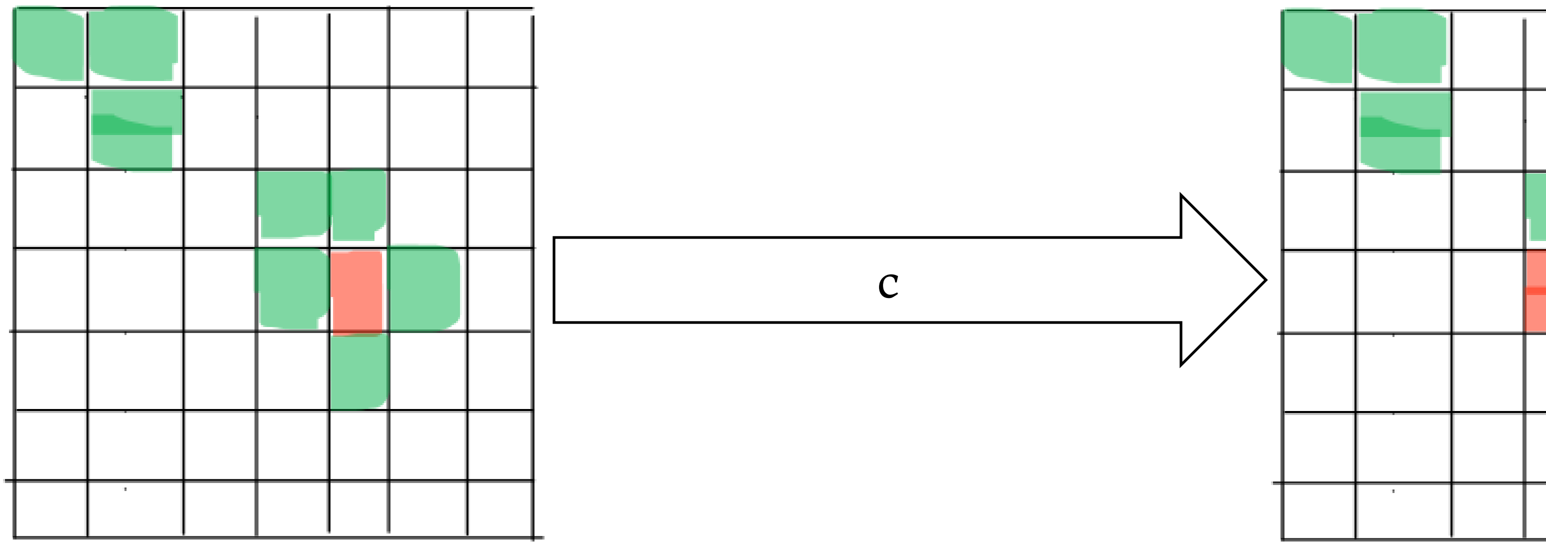
1. TREES GROW



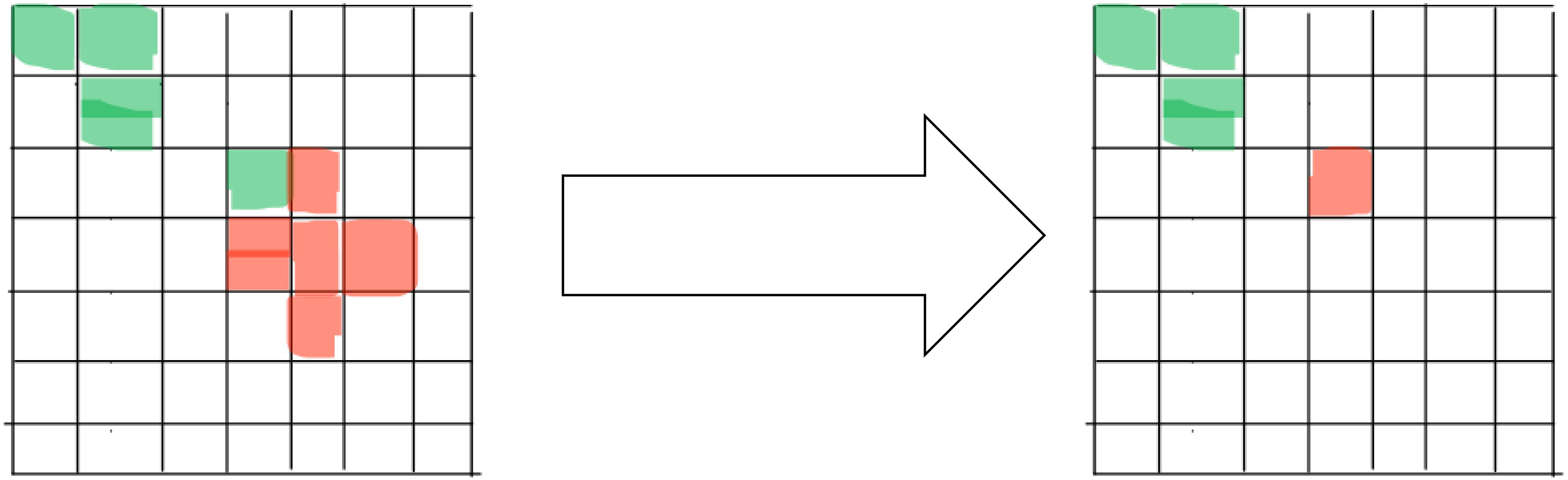
2. LIGHTNING STRIKES



3. THE FIRE SPREADS



4. THE TREE BURNS DOWN

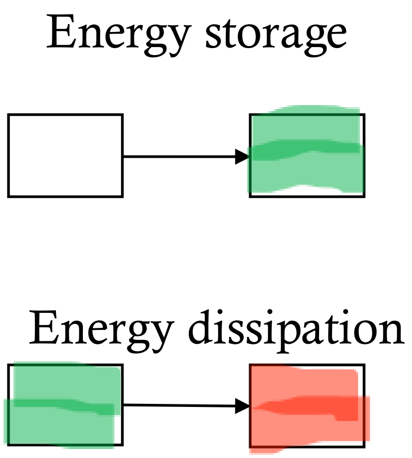


HOW WAS IT MADE?

- 1990 Per Bak, Kan Chen and Chao Tang introduced Model
  - Wanted to study spatial distribution of energy dissipation and storage

    →suspected fractal patterns in energy dissipation

- They proposed a critical point at p=0
  - As long as trees grow slowly, system is critical

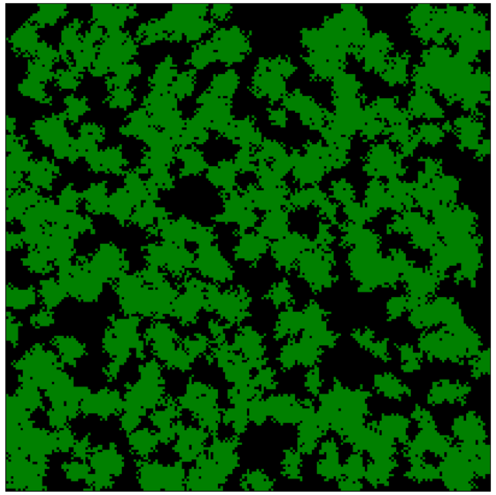


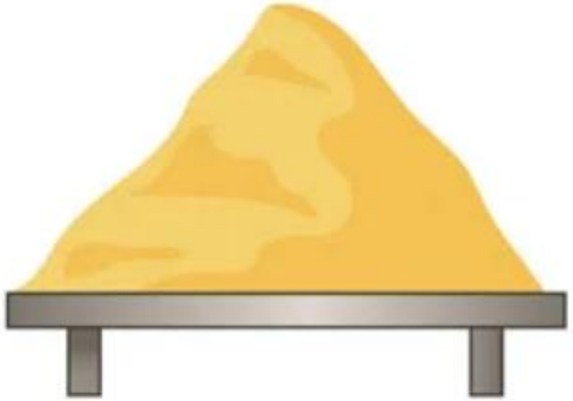
HOW WAS IT MADE?

- 1992: Drossel and Schwabl picked up the model
  - →argue against criticality of the system
  - →Introduce the lightning mechanism to induce criticality in the system
  - →Found a critical value

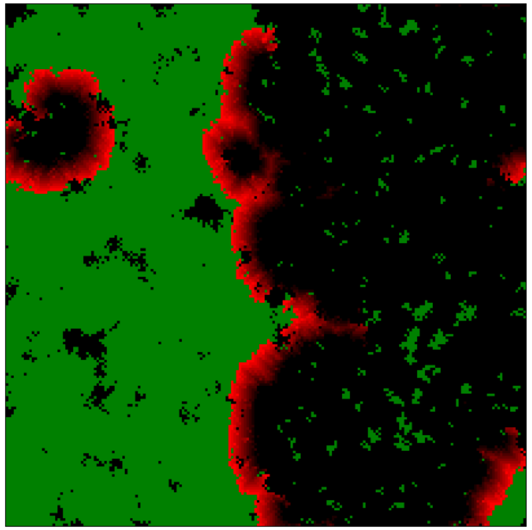
CRITICALITY

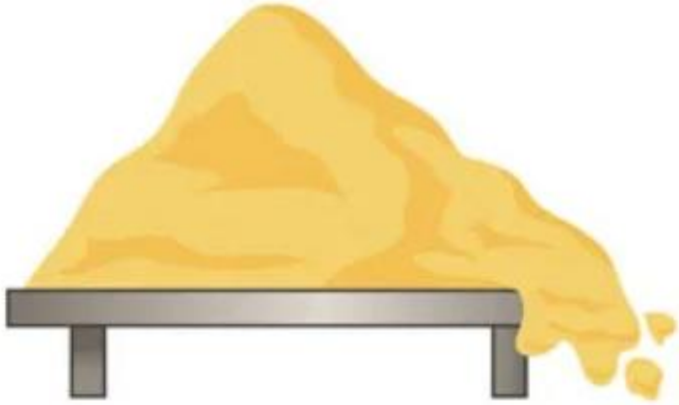
CRITICALITY – ENERGY STORAGE



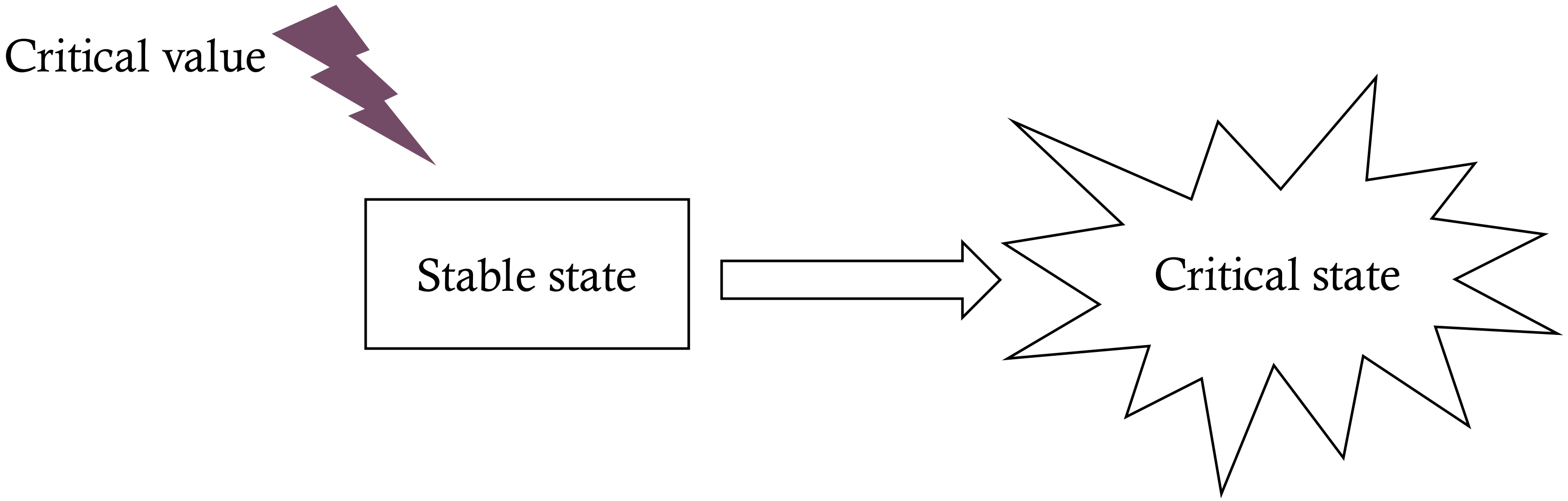


CRITICALITY- ENERGY DISSIPATION

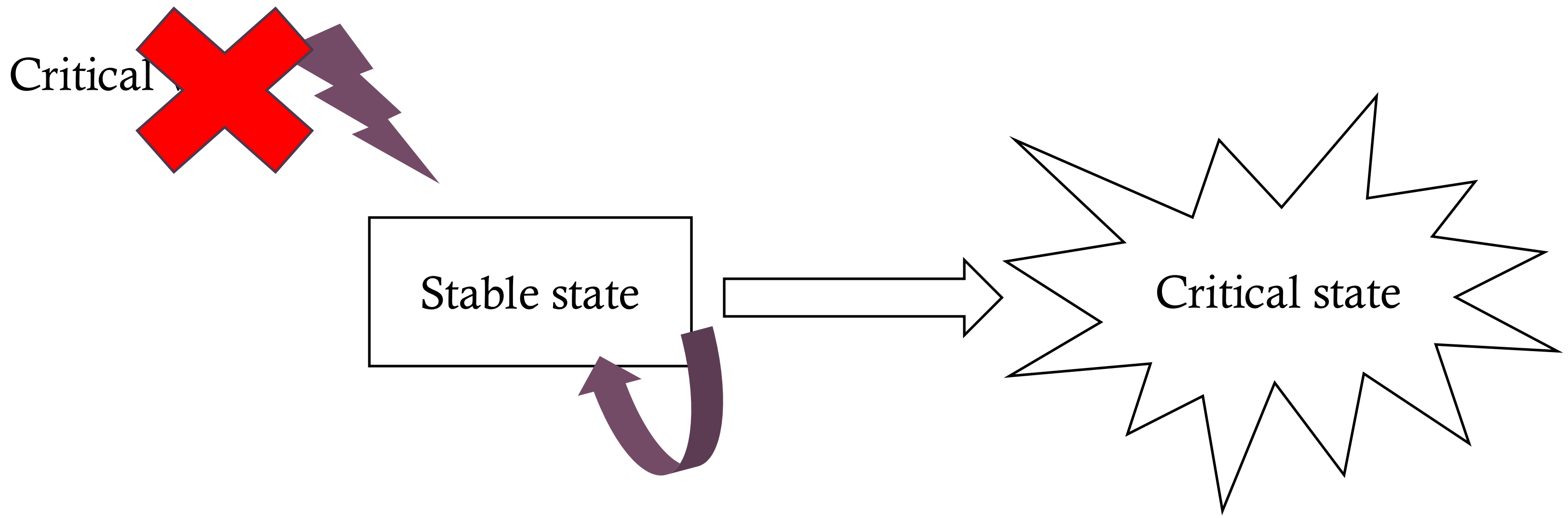




CRITICALITY-SOC



CRITICALITY- SOC



CRITICALITY-TIME SCALES

- Forest fire:

- sandpile:

$p<<c$

CRITICALITY-TIME SCALES

- Forest fire:

  $p \ll c$

  → Nothing happens

- sandpile:

  $p \ll c$

CRITICALITY-TIME SCALES

- Forest fire:

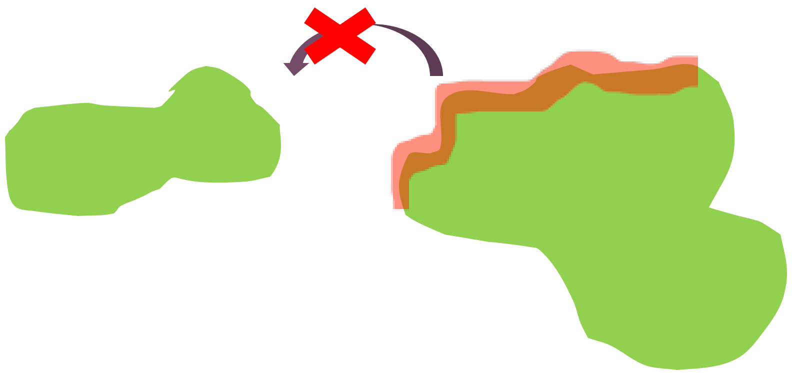
  $f<<p<<c$

  → Now it is critical

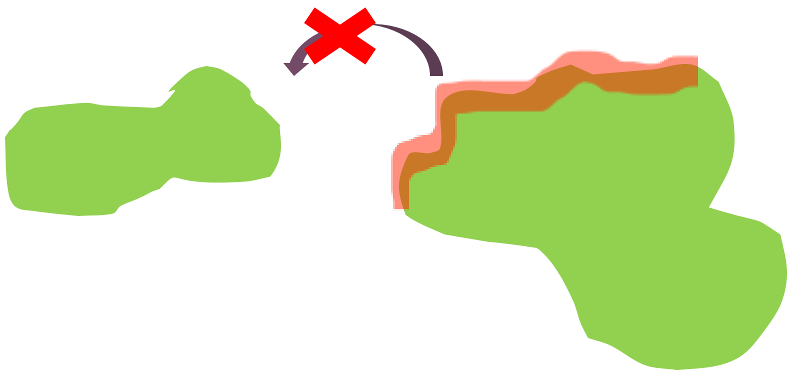
- sandpile:

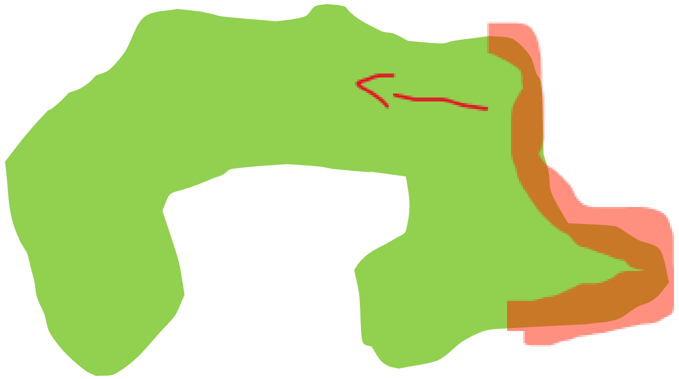
  $p<<c$

CRITICALITY-THE VALUE (?)



CRITICALITY- THE VALUE(?)





CRITICALITY – THE VALUE(?)

- → a lot of forest patches need to be connected in order for the fire to spread
- → the critical value lies in the <u>density</u> of the forest patches
- → or rather the f/p ratio

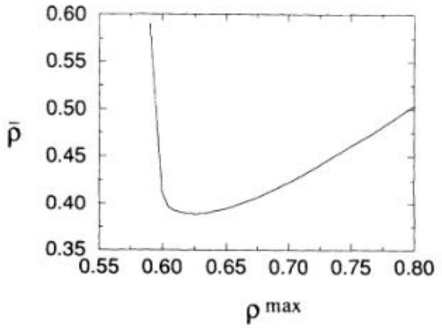


FIG. 3. Mean forest density as a function of maximum forest density for $d=2$.

- Critical state is reached if f/p→0
  - → the closer f/p ist to 0 the better you can observe the spiral patterns
- Determined a minimal forest density during critical state p=0,39

CRITICALITY- THE VALUE(?)

- https://itp.uni-frankfurt.de/~gros/StudentProjects/Projects_2020/projekt_lars_dingeldein/

WHAT IS IT USED FOR

- 1. studying self-organized criticality
- 2. also applicable for other scenarios:
  - →spreading of roumors
  - →spreading of diseases
- 3. not directly applicable for forest fires as it is

THANK YOU FOR YOUR ATTENTION

SOURCES

- P. Bak, K. Chen, and C. Tang, Phys. Lett. A 147, 297 (1990)
- B. Drossel and F. Schwabl, Phys. Rev. Lett. 69, 1629 (1992)
- P. Grassberger, New J. Phys. 4 ,17 (2002)
- https://itp.uni-frankfurt.de/~gros/StudentProjects/Projects_2020/projekt_lars_dingeldein/
- Plameri L and Jensen HJ (2020) The Forest Fore Model: The Subleties of Criticality and Scale Invariance. Front. Phys. 8:257. doi:10.3389/fphy.2020.00257
- https://researchfeatures.com/dislocation-avalanches-metal-deformation/